

Technologies Used: Azure Databricks | Apache Spark | Delta Lake | Power BI

PHASE 0 – Project Initiation & Understanding

The objective of this project is to design and implement an enterprise-grade data engineering pipeline that ingests raw data, processes it using the Medallion Architecture (Bronze, Silver, Gold), and delivers analytics-ready data for business intelligence reporting in Power BI.

Business questions addressed include:

- What is the daily and monthly revenue trend?
- Which regions contribute the most revenue?
- Which products are top performers?

Assumptions:

- Data arrives in batch mode as CSV files
- Data volume is moderate and scalable
- Data quality issues such as nulls and duplicates may exist

Success criteria:

- Reliable ingestion
- Clean and validated Silver data
- Analytics-ready Gold tables
- Successful Power BI dashboards

PHASE 1 – Architecture & Design

The project follows the Medallion Architecture:

Source → Bronze → Silver → Gold → Power BI

Roles of components:

- Azure Databricks: Data processing platform
- Apache Spark: Distributed data processing engine
- Delta Lake: ACID-compliant storage with versioning
- Power BI: Business intelligence and visualization tool

PHASE 2 – Source Data Preparation

Three datasets were created to simulate source systems:

- Sales Transactions
- Product Master
- Store / Region

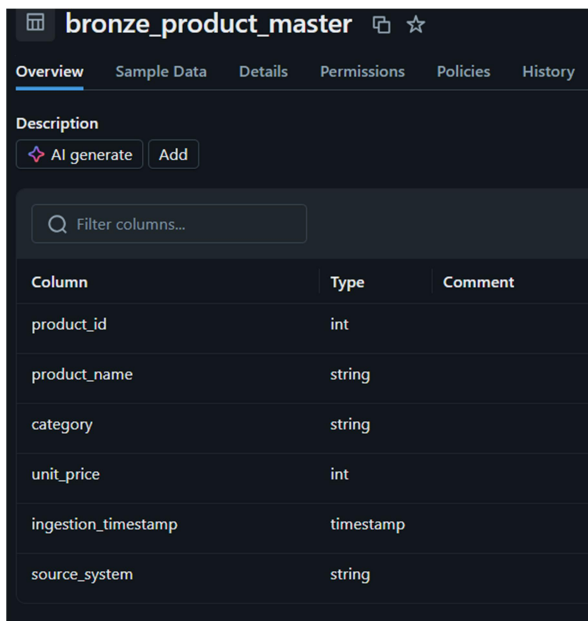
The datasets intentionally include null values, duplicates, and invalid references to test data quality and validation logic.

PHASE 3 – Bronze Layer (Raw Ingestion)

The Bronze layer stores raw data exactly as received from source systems. Minimal transformation is applied.

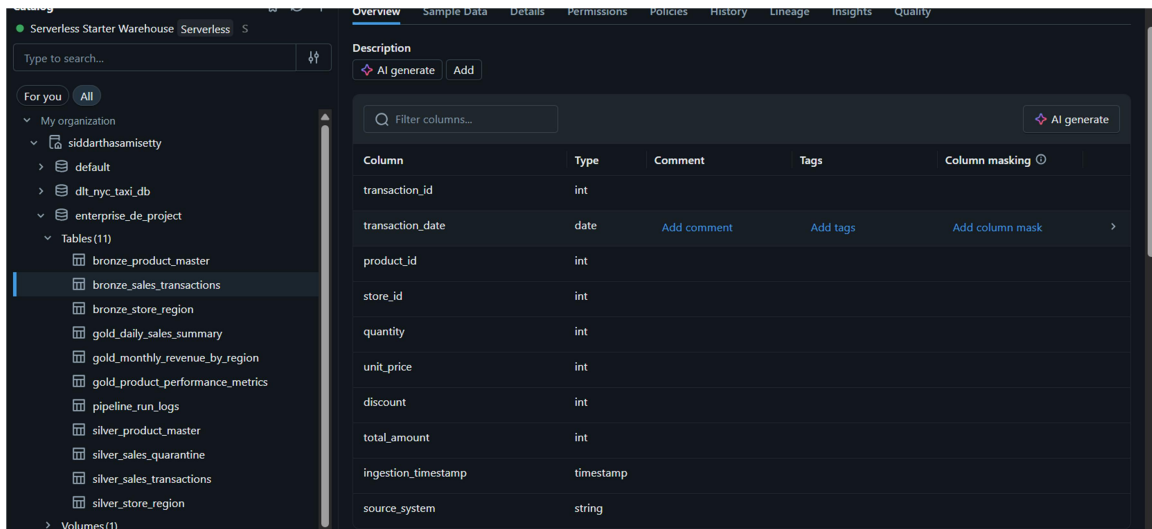
Key actions:

- Read CSV files using Spark
- Add ingestion metadata
- Store data as Delta tables



The screenshot shows a data catalog interface for a table named 'bronze_product_master'. The interface includes a header with the table name and icons for a grid, a copy icon, and a star icon. Below the header is a navigation bar with tabs: Overview, Sample Data, Details, Permissions, Policies, and History. The 'Overview' tab is selected. Under the 'Description' section, there are two buttons: 'AI generate' and 'Add'. Below this is a search bar labeled 'Filter columns...'. The main content is a table with the following columns: Column, Type, and Comment. The table contains six rows of data:

Column	Type	Comment
product_id	int	
product_name	string	
category	string	
unit_price	int	
ingestion_timestamp	timestamp	
source_system	string	



PHASE 4 – Silver Layer (Cleansing & Calibration)

The Silver layer applies data cleansing, validation, and business rule enforcement.

Key transformations:

- Remove duplicates
- Handle null values
- Validate foreign keys
- Recalculate total_amount using business logic

Invalid records are moved to a quarantine Delta table with rejection reasons.

PHASE 5 – Delta Lake Implementation

Delta Lake features were explicitly demonstrated:

- MERGE for incremental UPSERTs
- Schema evolution
- Time Travel
- Version rollback
- OPTIMIZE for performance

PHASE 6 – Gold Layer (Analytics-Ready Data)

The Gold layer contains business-level aggregations optimized for reporting.

Gold tables created:

- Daily Sales Summary
- Monthly Revenue by Region
- Product Performance Metrics

transaction_date	total_revenue	total_transactions
09 January 2024	2949590	28
14 January 2024	2078256	31
04 January 2024	2667068	31
27 January 2024	2638159	28
28 January 2024	2260840	28
08 January 2024	2301487	31
16 January 2024	2055714	25
12 January 2024	1831262	24
11 January 2024	1896990	25
03 January 2024	2515865	28
13 January 2024	2589881	27
30 January 2024	2031885	24
22 January 2024	2532441	29
18 January 2024	2172252	24
20 January 2024	2540390	24
15 January 2024	2522109	29
24 January 2024	2684890	28
25 January 2024	2249448	23
26 January 2024	2431712	26
19 January 2024	2620913	26
05 January 2024	2331747	25
17 January 2024	2844633	24
07 January 2024	1871932	24
06 January 2024	1958626	23
29 January 2024	1942315	25
01 January 2024	2258088	26
10 January 2024	2200614	28
03 January 2024	2600014	29

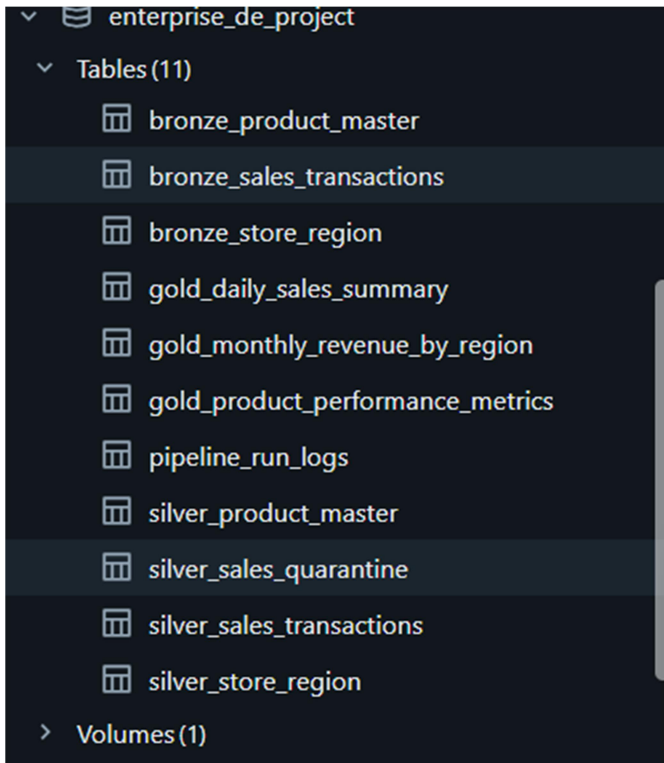
Table: gold_daily_sales_summary (30 rows)

year_month	region	monthly_revenue
2024-01	South	70051160
2024-03	West	72000
2024-03	South	71500
2024-02	South	149000
2024-02	North	60000

Table: gold_monthly_revenue_by_region (5 rows)

product_id	product_name	category	total_revenue	total_quantity_sold	total_transactions
122	Product_122	Electronics	795778	40	13
124	Product_124	Furniture	1239101	35	15
142	Product_142	Furniture	1798650	43	16
143	Product_143	Clothing	946719	30	12
148	Product_148	Clothing	1562317	59	17
130	Product_130	Clothing	1391870	49	17
144	Product_144	Clothing	1499742	46	15
114	Product_114	Furniture	1303000	44	13
104	Product_104	Clothing	1118380	40	13
113	Product_113	Clothing	1617533	53	17
126	Product_126	Furniture	585274	25	10
116	Product_116	Clothing	484871	26	10
115	Product_115	Clothing	871260	30	12
132	Product_132	Clothing	2002286	59	19
120	Product_120	Furniture	1624127	54	20
129	Product_129	Electronics	1855149	65	18
101	Product_101	Clothing	1434115	33	11
110	Product_110	Furniture	1912390	68	22
127	Product_127	Electronics	483119	12	5
117	Product_117	Clothing	1587104	57	17
141	Product_141	Clothing	566044	27	10
145	Product_145	Clothing	1802505	59	22
121	Product_121	Clothing	1531810	62	23
149	Product_149	Furniture	1590709	46	13
102	Product_102	Furniture	1807519	54	18
137	Product_137	Furniture	1439761	62	22
107	Product_107	Furniture	421784	27	10
160	Product_160	Electronics	8321843	36	16

Table: gold_product_performance_metrics (50 rows)

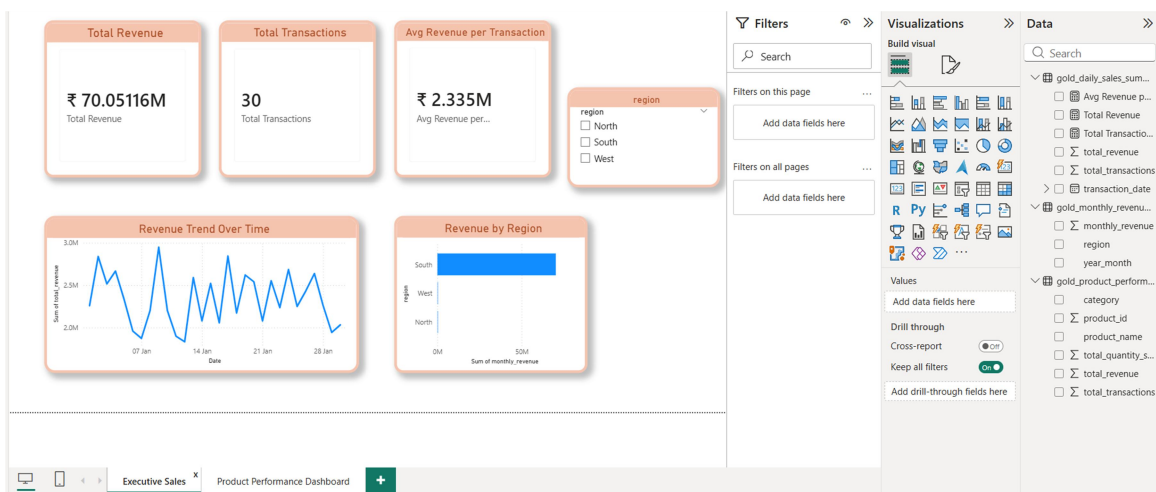


PHASE 7 – Power BI Integration

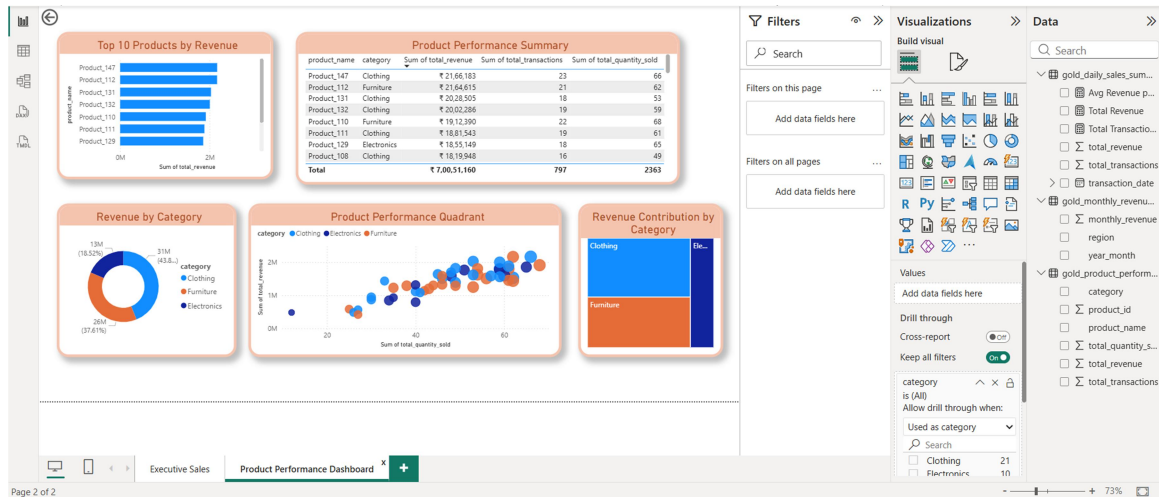
Power BI was connected directly to Gold Delta tables using the Azure Databricks connector.

Dashboards developed:

- Executive Sales Dashboard



- Product Performance Dashboard



PHASE 8 – Logging, Monitoring & Error Handling

A centralized Delta logging table (pipeline_run_logs) was created to capture:

- Execution start and end times
- Record counts per layer
- Quarantine counts
- Execution status and errors

The pipeline is idempotent and safe to rerun due to the use of Delta MERGE and overwrite logic.

Catalog Explorer > siddarthasamisetty > enterprise_de_project >

pipeline_run_logs [Open in a dashboard] [Share] [Create]

Overview | Sample Data | Details | Permissions | Policies | History | Lineage | Insights | Quality

Description

[AI generate] [Add]

Filter columns... [AI generate]

Column	Type	Comment	Tags	Column masking
pipeline_name	string			
layer	string			
step_name	string			
start_time	timestamp			
end_time	timestamp			
record_count	bigint			
status	string	Add comment	Add tags	Add column mask
error_message	string			

pipeline_name	layer	step_name	start_time	end_time	record_count	status	error_message
enterprise_de_pipeline	bronze	bronze_record_count	22/12/2025 13:39:24.927	22/12/2025 13:39:25.220	1000	SUCCESS	null
enterprise_de_pipeline	silver	silver_record_count	22/12/2025 13:39:24.927	22/12/2025 13:39:25.430	803	SUCCESS	null
enterprise_de_pipeline	gold	gold_record_count	22/12/2025 13:39:24.927	22/12/2025 13:39:25.652	30	SUCCESS	null
enterprise_de_pipeline	bronze	bronze_record_count	22/12/2025 13:42:06.230	22/12/2025 13:42:06.449	1000	SUCCESS	null
enterprise_de_pipeline	silver	silver_record_count	22/12/2025 13:42:06.230	22/12/2025 13:42:06.638	803	SUCCESS	null
enterprise_de_pipeline	gold	gold_record_count	22/12/2025 13:42:06.230	22/12/2025 13:42:06.825	30	SUCCESS	null
enterprise_de_pipeline	silver	silver_sales_processing	22/12/2025 13:42:13.583	22/12/2025 13:42:13.790	803	SUCCESS	null
enterprise_de_pipeline	silver	quarantine_record_count	22/12/2025 13:41:49.178	22/12/2025 13:41:49.178	203	SUCCESS	null
enterprise_de_pipeline	silver	silver_sales_processing	22/12/2025 13:38:59.266	22/12/2025 13:38:59.897	803	SUCCESS	null
enterprise_de_pipeline	silver	rerun_safety_validation	22/12/2025 13:41:25.071	22/12/2025 13:41:29.127	803	SUCCESS	null
enterprise_de_pipeline	silver	quarantine_record_count	22/12/2025 13:39:56.589	22/12/2025 13:39:56.589	203	SUCCESS	null